Candidate Seat No:________________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Computer Applications (MCA)
Semester-II University Examination May-2018
CA 729 Operating System Concepts and Network Technology
Date: 09.05.18, Wednesday   Time: 10.00 a.m. to 01.00 p.m.   Maximum Marks: 70

Instructions:

1. **Attempt both sections in separate answer books.**
2. **The figure to the right indicates marks.**
3. **Make a suitable assumption when necessary.**

SECTION-1

Q.1 **Give an appropriate answer from following multi choice questions**. **[11]**

1. How many states of processes are part of main memory in seven state process model?
(a) 1 **(b)** 2 **(c)** 3 **(d)** None of these

2. __________concept of memory management is not visible to the user.
(a) Paging **(b)** Segmentation **(c)** (a) and (b) both **(d)** None of these

3. The listing of the sequence of the instructions that is executed for the process is termed as ___________
(a) Process Track **(b)** Process Trace
(c) Program Track **(d)** Process Block

4. If there are no ready processes and main memory is full___________ transition occurs according to seven state process model.
(a) Blocked→Blocked/Suspend **(b)** Blocked/Suspend→Ready/Suspend
(c) Ready/Suspend→ Ready **(d)** Ready→ Ready/Suspend

5. ______section is the section of code within a process that requires access to shared resources.
(a) Semaphore **(b)** Critical **(c)** Bad **(d)** Memory

6. semWait() operation of semaphore __________the semaphore value.
(a) Initialize **(b)** assigns **(c)** decrement **(d)** increment

7. __________scheduling policy is non preemptive policy.
(a) FCFS **(b)** SPN **(c)** SRN **(d)** (a) and (b) both

8. An __________of program running on a computer is termed as the process.
(a) Interface **(b)** Instance **(c)** Inquiry **(d)** Interrupt

9. _________placement policy is an ideal policy.
(a) First Fit **(b)** Next Fit **(c)** Best Fit **(d)** None of these

10. ____________mutual exclusion technique coordinates the processes by means of signaling.
(a) Monitor **(b)** Message Passing **(c)** Critical Evaluation **(d)** Semaphore

11. External fragmentation of dynamic partitioning can be removed with _____________technique.
(a) Coding **(b)** Controling (**c**) Compaction (**d**) None of these

[**P.T.O**]

Q.2 [A] What do you mean by Operating System? Explain five functions of it. **[06]**

[B] Explain five state process model in detalis. **[06]**

OR

Q.2 [A] What are the characteristics of RAID model? Explain all its layers in details . **[06]**

[B] Give the operations of semaphore and provide the solution of the producer-consumer problem using it. **[06]**

Q.3 **Give answers in detail (Any Three)**. **[12]**

1. Explain two state process model with all aspects.

2.Explain First Come First Serve(FCFS) scheduling policy with appropriate example.

3. What do you mean by Deadlock? Exaplain conditions of it.

4. Give the solution of the Dining Philosopher problem.

5. Explain logical and physical organization requirements of memory management.

SECTION-2

Q.4 [A] **Match column A with the correct answer on column B.** **[05]**

Column A	**Column B**
1. Presentation Layer	a. Mail Service
2. Application Layer	b. RTP
3. Network Layers	c. TCP
4. Session Layer	d. Compression
5. Transport Layer	e. Packets

[B] **Give an appropriate answer from following multi choice questions**. **[06]**

1. IPv4 addresses are __________ in length.

(a) 32 - bit **(b)** 64 - bit **(c)** 16 - bit **(d)** None of the above

2. ________is the protocol data unit of the transport layer.

(a) Segment **(b)** Frame **(c)** Bits **(d)** None of the above

3. What is the default subnet mask of Class C network?

(a) 255.0.0.0 **(b)** 255.255.0.0 **(c)** 255.255.255.0 **(d)** None of these

4. Internet Protocol Security(IPSec) is an _____________ standard suite of protocols.

(a) Internet Engineering Task Force
(b) Internet Engineering Task Field
(c) International Engineering Task Force
(d) Internet Establishment Task Force

5. A ______ routing table is created, maintained, and updated by a network administrator, manually

(a) Dynamic **(b)** Static **(c)** Both **(d)** None of these

6. The processing power of computer system required to run asymmetric algorithm is _______.

(a) Less **(b)** Higher **(c)** Same **(d)** Can’t say

Q.5[A]	Explain Symmetric Key Encryption using salient features.	**[06]**
[B]	Explain Distance Vector Routing algorithm with example.	**[06]**
	OR	
Q.5[A]	Explain Physical layer of OSI model with all major functions.	**[06]**
[B]	What do you mean by network security? Explain any five types of network security.	**[06]**
Q.6	**Give answers in detail (Any Three)**.	**[12]**

1. Explain any four functions of transport layer of OSI model.
2. Explain the operations of the Kerberos protocol .
3. Explain IPSec standard for cryptography.
4. What do you mean by subnet mask? Give the default subnet mask of all Classes with network bits, host bits and total hosts.
5. Explain "SSL Handshake" Process.
